LMRA Problem – 6 **Date : 06-04-2026**

Fit a logistic regression model to the **logistic_regression dataset** considering **Passed** as the response variable and **Studied** and **Slept** as predictor variables. Find the **odds ratios** and perform the **Likelihood Ratio Test** to check the overall significance of the model.

Solution :

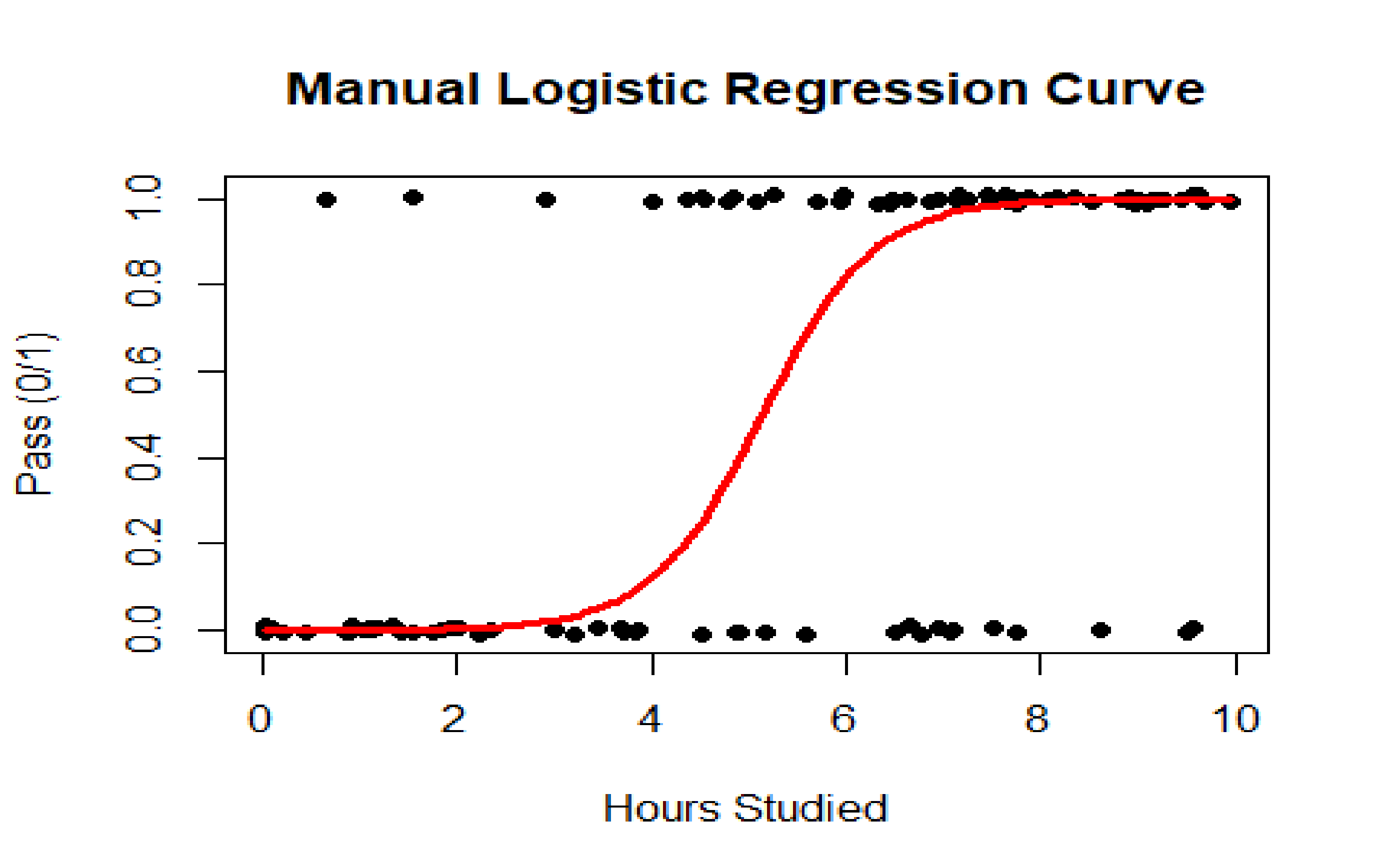
Source Code :

Manually :

```
library(MASS)
data<-read.csv("D:/REGRESSION CLASS/logistic_regression.csv")

#manually

y<-as.matrix(data$Passed)
x1<-data$Studied
x2<-data$Slept

#Design matrix ( intercept + predictor)
x<-as.matrix(cbind(1,x1,x2))

#intialize Beta value

Beta<-matrix(0,ncol = 1,nrow =ncol(x))

#iteration loop for eta

for(i in 1:100){
  #step 1: linear predictor
  eta<-x%*%Beta

  #step 2: probability
  p<-exp(eta)/(1+exp(eta))

  #step 3: weight matrix
  W<-diag(as.vector(p*(1-p)))

  #step 4: Score Vector
  score<-t(x)%*%(y-p)

  #step 5: Hessian Matrix
  h<-t(x)%*%W%*%x

  #estimated value
  beta_new<-Beta + ginv(h)%*%score

  #control the loop
  if(max(abs(beta_new - Beta))<0.0001)break

  #reassign the Beta
  Beta=beta_new

}
Beta                                             #Logistic regression Model

##             [,1]
## [1,] -16.160247
## [2,]   1.745379
## [3,]   1.486506

#odds ratio expectation

odd_ratio<-matrix(data=exp(Beta),ncol=3)
odd_ratio

##              [,1]     [,2]    [,3]
## [1,] 9.58725e-08 5.728073 4.42162
```

```r
#likelihood ratio test

full_model<-glm(y ~ x1+x2,family = "binomial")
full_model

##
## Call:  glm(formula = y ~ x1 + x2, family = "binomial")
##
## Coefficients:
## (Intercept)           x1           x2
##     -16.160        1.745        1.487
##
## Degrees of Freedom: 99 Total (i.e. Null);  97 Residual
## Null Deviance:       137.6
## Residual Deviance: 43.4  AIC: 49.4

reduced_model<-glm(y ~ 1 , family = "binomial")
reduced_model

##
## Call:  glm(formula = y ~ 1, family = "binomial")
##
## Coefficients:
## (Intercept)
##      0.2007
##
## Degrees of Freedom: 99 Total (i.e. Null);  99 Residual
## Null Deviance:       137.6
## Residual Deviance: 137.6     AIC: 139.6

#loglikelihood of both models

ll_full<-as.numeric(logLik(full_model))
ll_redu<-as.numeric(logLik(reduced_model))

#Loglikelihood ratio(LRT)

LRT<-2*(ll_full - ll_redu)
LRT

## [1] 94.22618

#degree of freedom

df<-attr(logLik(full_model),"df") - attr(logLik(reduced_model),"df")

p_value<-1 - pchisq(LRT,df)
p_value

## [1] 0

#plotting the Logistic regression curve

x2_mean<-mean(x2)

studied_seq<-seq(min(x1),max(x1),length=100)

X_new<-cbind(1,studied_seq,x2_mean)

eta_curve<-X_new%*%Beta

p_curve<-exp(eta_curve)/(1+exp(eta_curve))

y<-data$Passed                        #response variable
x1<-data$Studied                      #1st predictor variable
x2<-data$Slept                        #2nd predictor variable

#logistic regression model

model<-glm(y ~ x1+x2,family = binomial)

odds_ratio<-exp(coef(model))
```

```r
#likelihood ratio test

anova(model,test = "Chisq")

Hypothesis

if(p_value < 0.05){
  print("On the basis of given data we Rejected the NUll Hypothesis")
}else{
  print("On the basis of given data we fail to reject the Null Hypothesis")
}

## [1] "On the basis of given data we Rejected the NUll Hypothesis"

plot(x1, jitter(y,0.05),
     pch=16,
     xlab="Hours Studied",
     ylab="Pass (0/1)",
     main="Manual Logistic Regression Curve")

lines(studied_seq,p_curve,lwd=3,col="red")
```



🎓 **Interpretation :**

- Hours studied and hours slept significantly affect the probability of passing the exam.
- As study time increases, the chance of passing increases following an S-shaped logistic curve.